

32. Morphological operations based on OpenCV using Closing technique.

AIM:

Morphological operations using Closing technique.

PROGRAM:

```
import cv2
import numpy as np

img = cv2.imread(
    r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png",
    cv2.IMREAD_GRAYSCALE
)

if img is None:
    print("Error: Image not found!") else:
    # Create a 5x5 kernel    kernel =
    np.ones((5, 5), np.uint8)

    # Apply Closing operation (Dilation followed by Erosion)
    closing = cv2.morphologyEx(img, cv2.MORPH_CLOSE, kernel)

    # Save the output image    cv2.imwrite(
    r"C:\Users\Susmitha\Downloads\ITA0510 Lab\Closing.jpg",
    closing
    )

    # Display the images
    cv2.imshow("Original Image", img)
    cv2.imshow("Closing Image", closing)

    cv2.waitKey(0)
    cv2.destroyAllWindows() INPUT:
```



OUTPUT:



RESULT:

Thus the program is executed successfully.